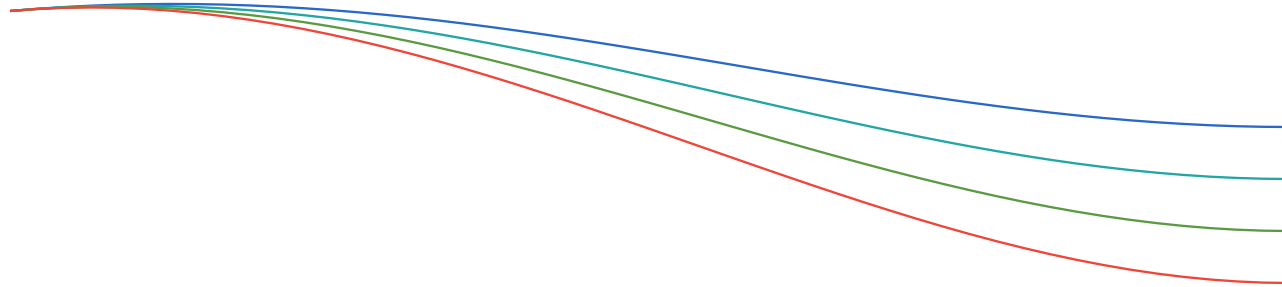


Constrained Optimal Glide Paths for DC Pensions

“Four Mean–Variance Solution Concepts” × “Strict / Clip” × “Rolling Evaluation”



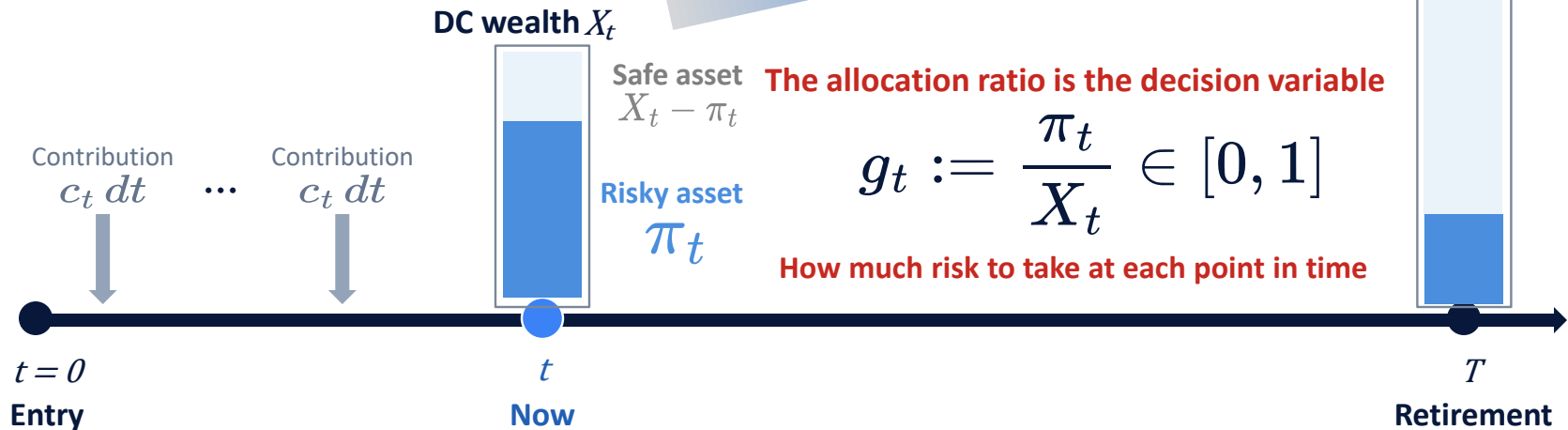
Yoshiki Nagayama

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Maximize retirement wealth by optimizing the allocation ratio

$\max_{g \in [0,1]}$

Objective J_t
 $= E_t[X_T] - \kappa * Var_t[X_T]$



The objective and the constraint on π determine the optimal glide path

Adding c , π , and an investment constraint to the Black–Scholes market yields a DC wealth model X

① Black–Scholes market

- Safe asset B_t

$$\frac{dB_t}{B_t} = r dt$$

- Risky asset S_t

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

② DC-specific inputs

$$+ c_t dt \quad \text{Contributions}$$

*DB benefits can also be incorporated

$$+ \pi_t \quad \text{Investment risky asset}$$

③ DC wealth X_t

$$dX_t = c_t dt + \pi_t \frac{dS_t}{S_t} + (X_t - \pi_t) \frac{dB_t}{B_t} = (rX_t + c_t + \beta\pi_t) dt + \sigma\pi_t dW_t$$

The investable amount cannot exceed current DC wealth

$$0 \leq \pi_t \leq X_t$$

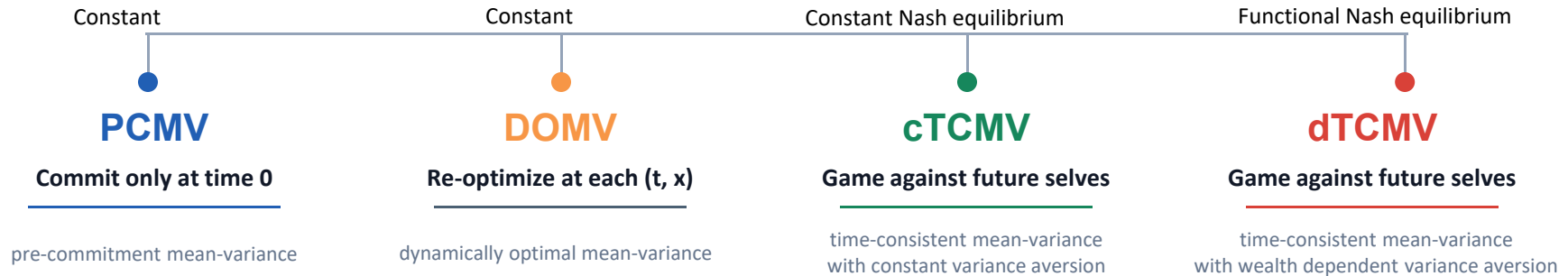
The market, contributions, and investment constraint determine DC wealth dynamics

Mean–variance optimization has multiple solution concepts

Common objective-function framework

$$\max_{\pi \in [0, x]} E_t[X_T | X_t = x] - \kappa(t, x) \text{Var}_t[X_T | X_t = x]$$

Definition of max



This study tests four optimality concepts under the same model and constraint

Reference: Mean–variance optimization formulations

Solution concept	Decision principle	Core criterion	Optimization method	Unconstrained solution ($y = x + H$)
PCMV	Commit at time 0	$E_0[X_T X_0 = x] - \kappa_P * Var_0[X_T X_0 = x]$	HJB equation	$\frac{\beta}{\sigma^2} \{ m e^{-r(T-t)} - y \}$
DOMV	Re-optimize at each (t, x)	$E_t[X_T X_t = x] - \lambda_D * Var_t[X_T X_t = x]$	HJB equation	$\frac{\beta}{2\lambda_D\sigma^2} e^{(A-r)(T-t)}$
cTCMV	Game against future selves (fixed risk aversion)	$E_t[X_T X_t = x] - \gamma_c * Var_t[X_T X_t = x]$	Extended HJB system (Nash equilibrium)	$\frac{\beta}{\gamma_c\sigma^2} e^{-r(T-t)}$
dTCMV	Game against future selves (state-dependent risk aversion)	$E_t[X_T X_t = x] - \frac{\rho_d}{x + H_t} * Var_t[X_T X_t = x]$	Extended HJB system (Nash equilibrium)	$\theta_{\rho_d}(t)y$

Adding DC-specific conditions to prior comparisons and numerically testing clip–strict gaps

Prior research

van Staden et al. (2021)

Compare exact forms of four concepts by terminal mean

PCMV DOMV cTCMV dTCMV

terminal distribution comparison



Four additional conditions examined here

Hidden wealth H = risk capacity from the present value of future contributions

Investment constraint $0 \leq \pi \leq X$

Strict–clip approximation error

Rolling evaluation of stochastic outcomes beyond time 0



CENTRAL RESEARCH QUESTION

What is optimal DC investment?

Can an easily generated clip solution replace a computationally intensive strict solution?

This study seeks policies that balance optimality and practical feasibility

Main Result 1 Unified clip/strict solution for four concepts

Clip solves then truncates; strict solves within the constraint

clip | low-cost approximation

- 1 The $Y = X + H$ transform makes the unconstrained problem easy to solve

$$\hat{\pi}^{unc}(t, x + H_t)$$



- 2 Solve the unconstrained PDE, then project onto $[0, X]$

$$\hat{\pi}^{clip}(t, x) = \text{Proj}_{[0, x]} \{ \hat{\pi}^{unc}(t, x) \}$$

$\neq$

Practicality © Optimality ×

strict | guaranteed optimality

- 1 Solve the constrained PDE

$$\tilde{\pi}(t, x) = -\frac{\beta}{\sigma^2} \frac{v_x}{v_{xx}}$$



$$v_t^m + \inf_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)v_x^m + \frac{1}{2}\sigma^2\pi^2 v_{xx}^m \right\} = 0,$$

- 2 Endogenize $[0, X]$ at each state while solving the constrained PDE

$$\pi^{strict}(t, x) = \text{Proj}_{[0, x]} \{ \tilde{\pi}(t, x) \}$$

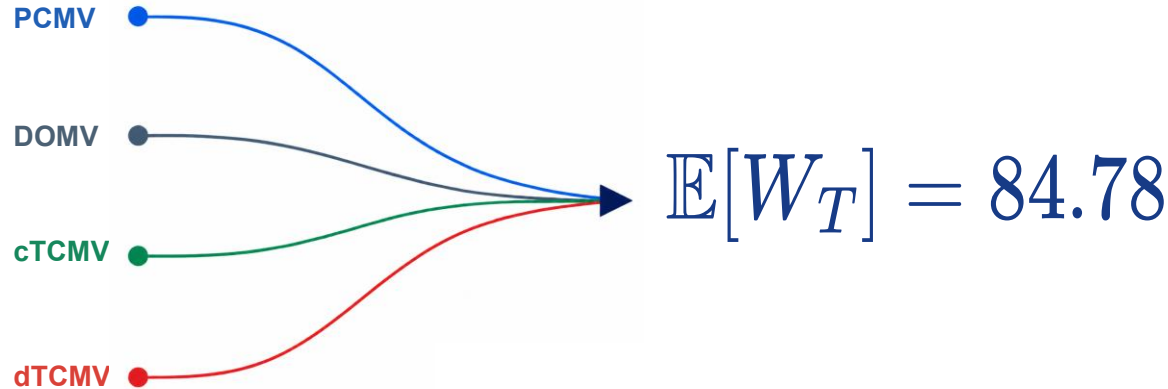


Practicality △ Optimality ©

Even with the same projection set, embedding the constraint in the PDE changes both meaning and tractability

Matching expected terminal wealth at 84.78 isolates distribution shape and risk timing

Calibrate each objective coefficient so all four strategies have the same mean

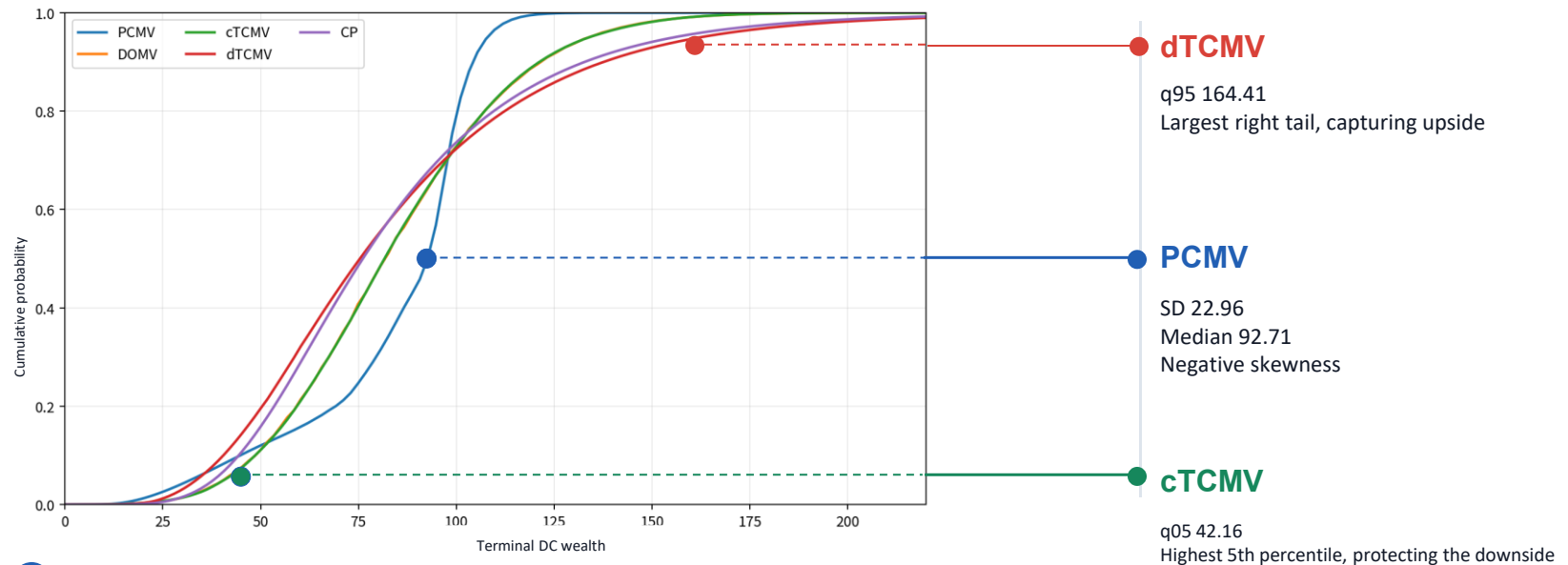


Common monthly assumptions over 40 years

Risk-free rate r	1.5%
Expected equity return μ	5.5%
Equity volatility σ	18%
Investment horizon T	40 years = 480 periods
Annual contribution	1
Initial wealth	1/12

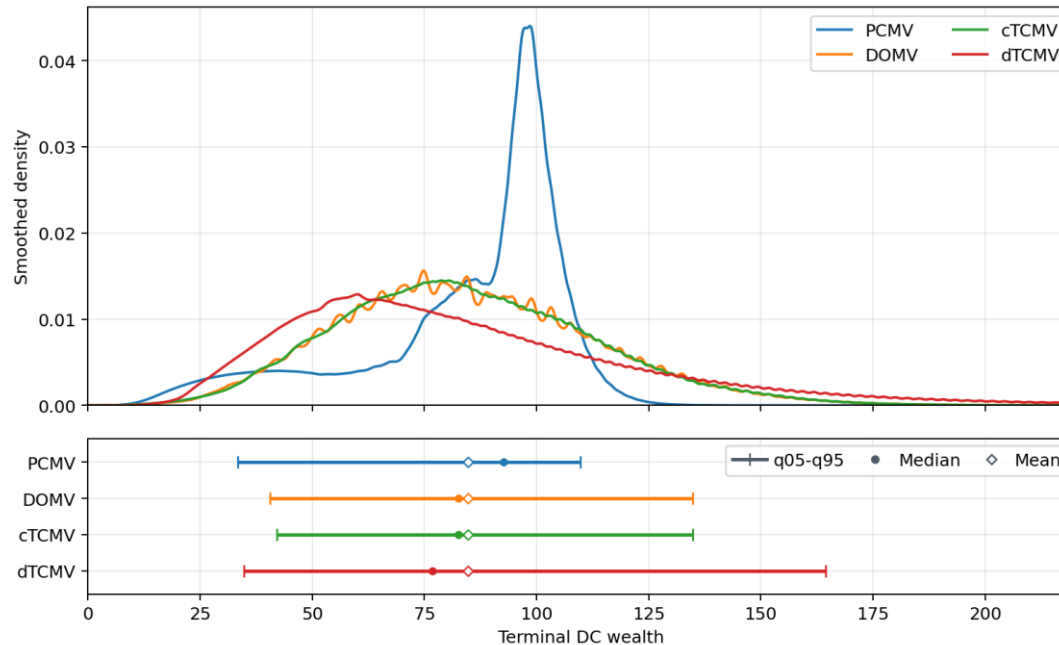
Removing mean differences enables a fair comparison of distribution shape and risk timing

With the same mean, left tail, median, and right tail differ substantially



➔ The same mean can produce very different distributions

Reference: Wealth distribution at retirement



dTCMV

q95 164.41

Largest right tail, capturing upside

PCMV

SD 22.96

Median 92.71

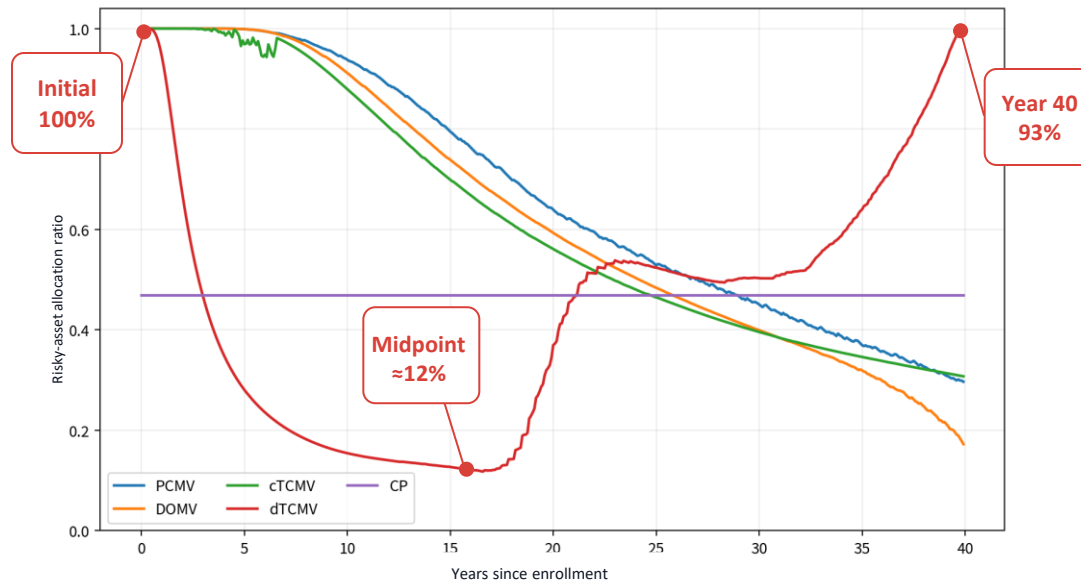
Negative skewness

cTCMV

q05 42.16

Highest 5th percentile, protecting the downside

dTCMV is U-shaped; other strategies are broadly monotone decreasing



Competing forces behind the U-shape

glide path = $\theta \times (1 + H/X)$

Background-wealth ratio H/X
↓

Equilibrium coefficient θ ↑

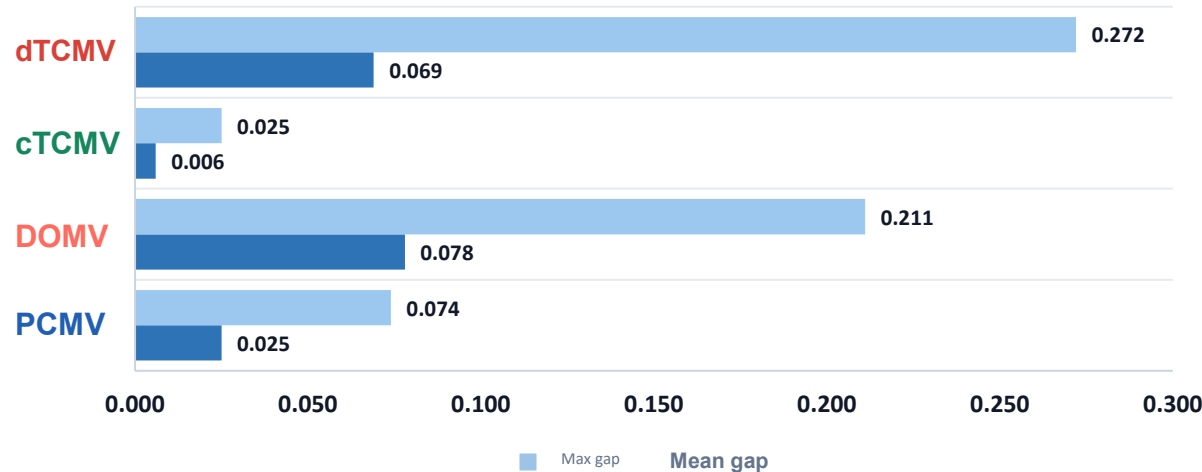
Later: θ
dominates



An age-only downward curve
cannot capture this state
dependence

Clip approximation accuracy depends on the solution concept

Glide path



cTCMV | mean gap 0.006

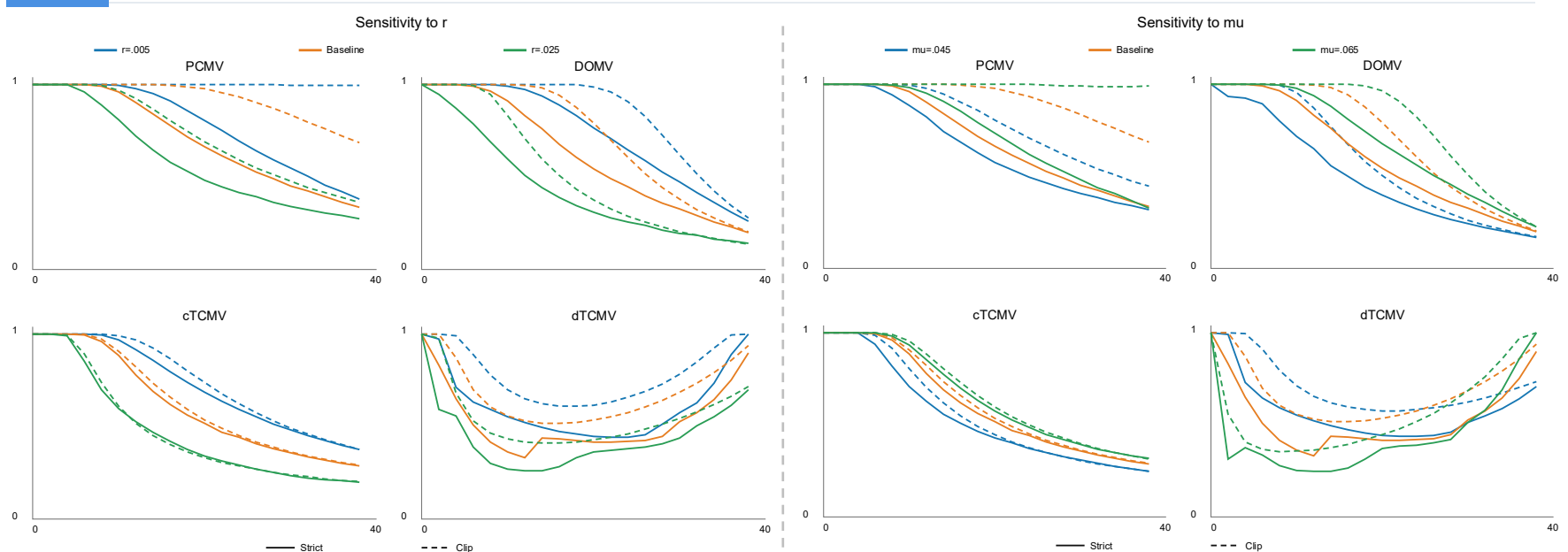
The easy-to-compute clip solution closely approximates the strict solution

DOMV | mean gap 0.078

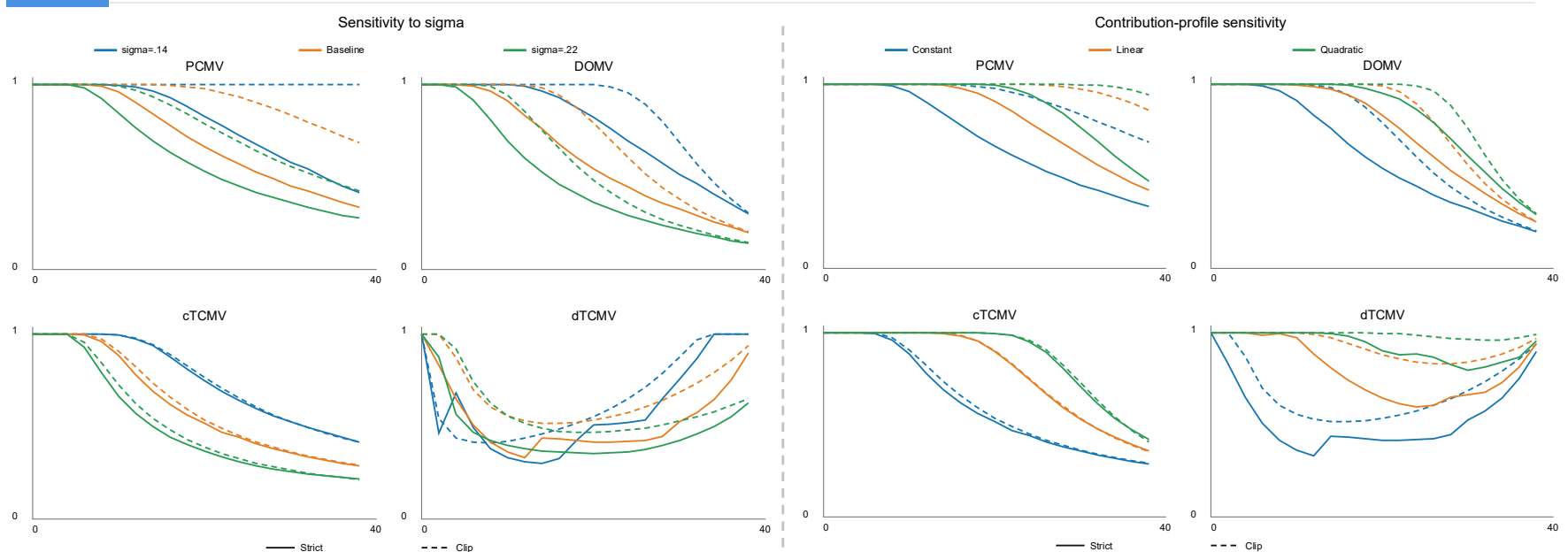
dTCMV | mean gap 0.069

The easy-to-compute clip solution does not reproduce the strict solution

Reference: Sensitivity to the risk-free rate r and risky-asset return μ

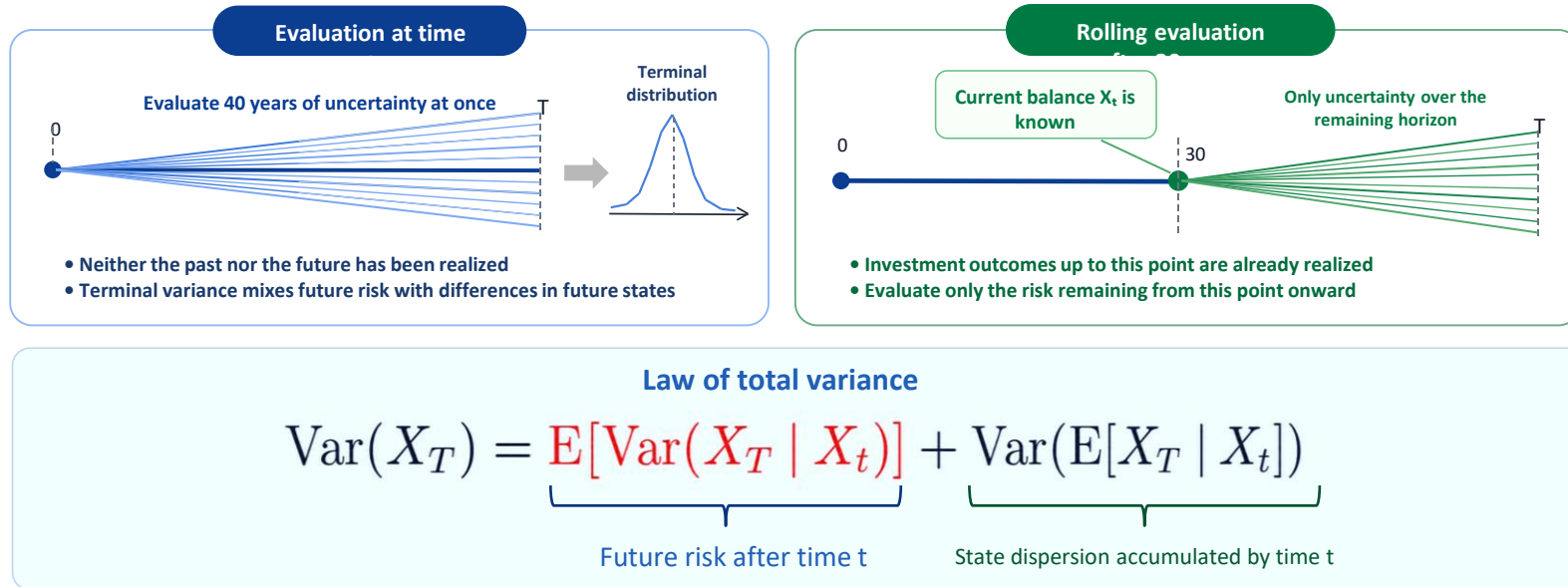


Reference: Sensitivity to risky-asset volatility σ and contribution c



Main Result 2 M-G-H Engine & Rolling Evaluation

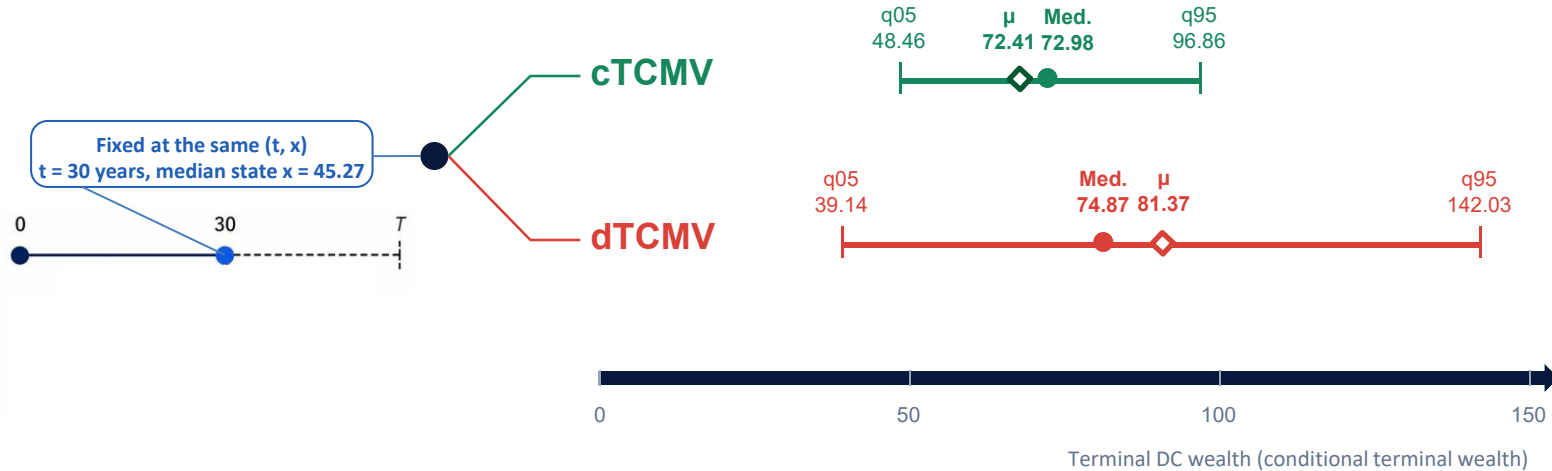
Why rolling evaluation requires risk reassessment at each date



➔ Conditioning on interim wealth enables a new risk measure driven by wealth variation during accumulation

Main Result 2 M-G-H Engine & Rolling Evaluation

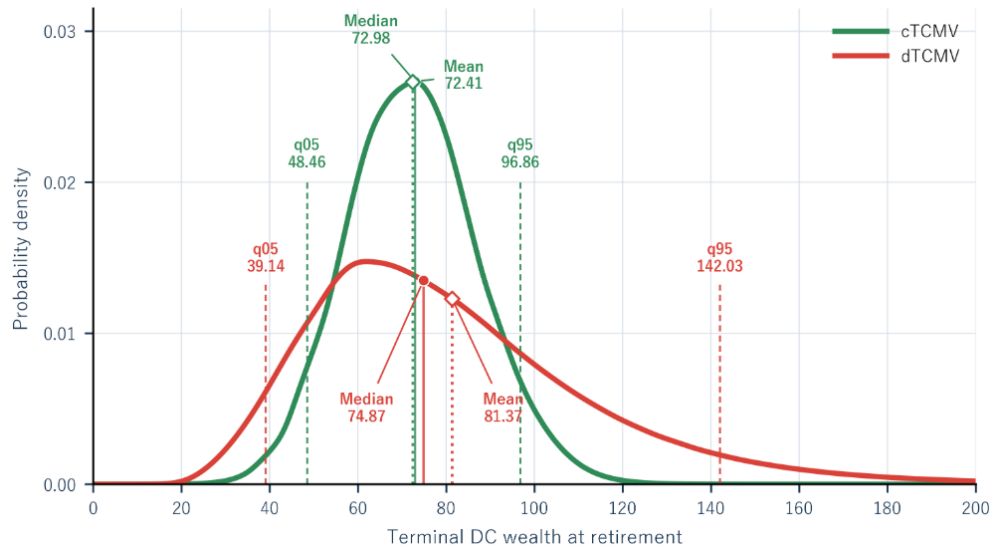
Starting from a future balance, dTCMV develops a wider upper tail



→ dTCMV widens the upside substantially, while q05 is slightly below cTCMV

Reference: Wealth distribution at retirement after rolling evaluation

Rolling evaluation | $t = 30$ years, median state $x = 45.27$



Solution concept	mean	q05	median	q95
cTCMV	72.41	48.46	72.98	96.86
dTCMV	81.37	39.14	74.87	142.03



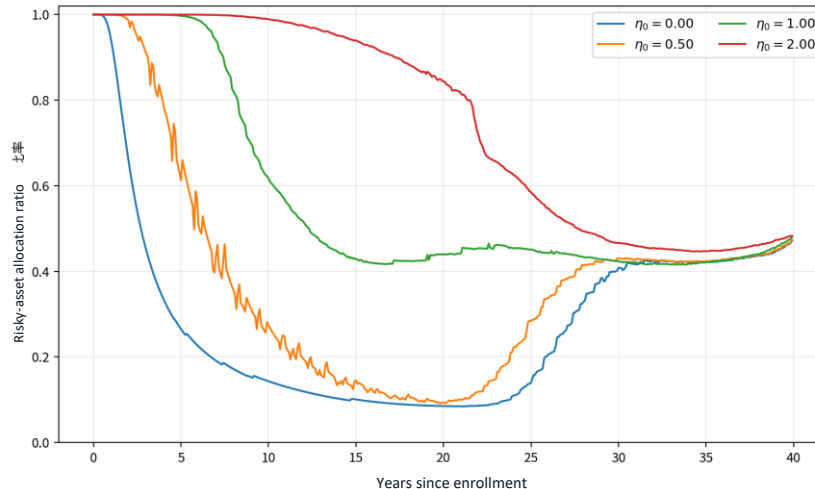
Changing future policy reveals the trade-off between upside participation and downside risk

Theory in progress: simulation only

Skewness preference reshapes when risk is taken

$$J_{MVS} = \mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T) + \frac{\eta}{3} \mathbb{E}[(W_T - \mathbb{E}[W_T])^3]$$

Add skewness
to dTCMV
(dTCMVS)



Evolution under dTCMVS

- ✓ Clear glide-path shift
- ✓ Retirement distribution: larger upper tail, slightly higher lower-tail risk

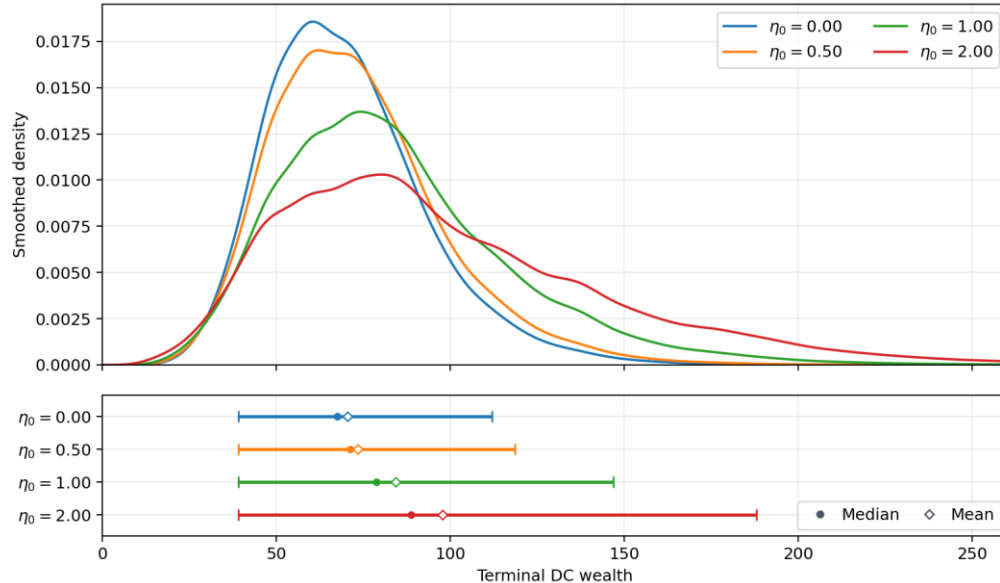
Other confirmed findings

- ✓ $\eta = 0$ recovers MV
- ✓ Validated for all four strategies

➔ Positive η controls risk timing toward upside and a more conventional shape

Theory in progress: simulation only

Reference: Retirement wealth distribution under MVS optimization



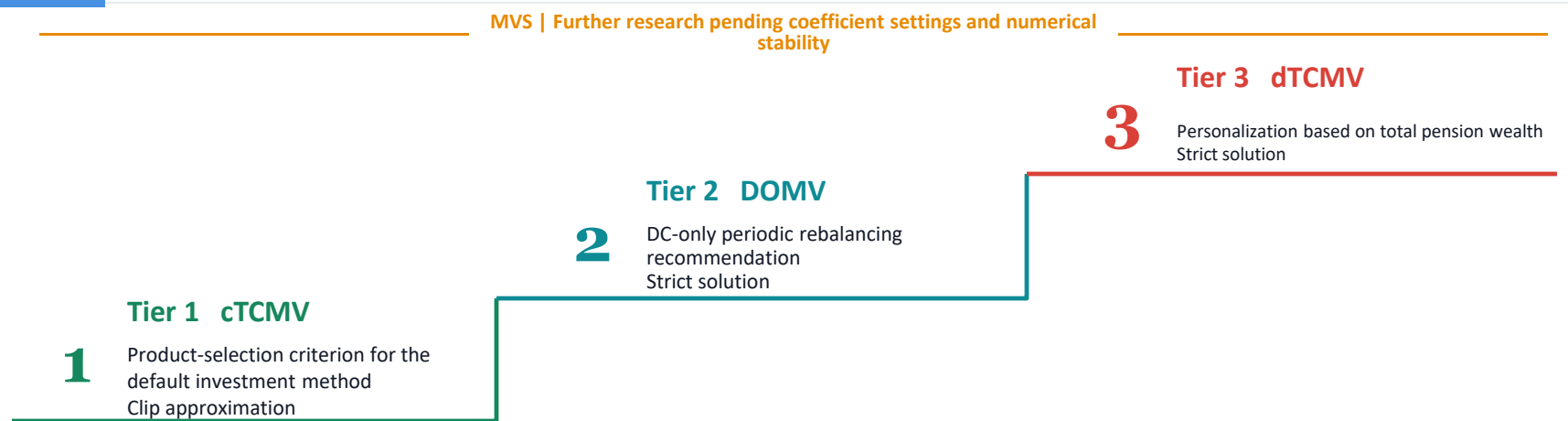
η	Mean	q05	median	q95
0.00 (MV)	77.65	47.16	74.72	118.15
2.00 (MVS)	104.24	41.98	89.21	217.53



Skewness preference generated substantial upside

Main Result 3 Translating solution concepts into a three-tier operating model

Four solution concepts translate into a practical three-tier operating model



Summary: Contributions and practical implications

01

Endogenous glide paths under DC constraints

Declining TDF-like and U-shaped paths can emerge endogenously from different decision principles.
The results support the conventional declining glide path as a valid outcome of optimisation in a simple DC setting.

02

Simple clipping is not always enough

Clipping works well for cTCMV, but direct constraints matter for some solution concepts.

03

Evaluate and manage risk from the member's current state

Rolling evaluation compares future outcomes from the same balance and supports three-layer design.